



EVALUATION

BY NELSON OGODO



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Assets and Content

How appropriate were assets used

The assets for Greenfield Local Hub prototype was intentionally chosen for the intended purpose of the platform and its target audience. The target audience for Greenfield local hub includes families, local community members and people who desire fresh, locally sourced food. So considering all this, the product images chosen from sources such as Pixabay, Freepik and iStock were chosen to be high quality and actual representations of the actual products that would be sold which included organic honey, free range eggs, fresh milk, carrots, strawberries and lettuce.

The images successfully supported the purpose of the platform which is to build trust and transparency with customers. When users were asked in the feedback from if the product images represented actual products, 100% of users agreed that the images were successful. This is important because the business proposal clearly highlighted transparency a key goal for GLH, noting that being transparent about pricing builds loyalty with customers so realistic images also directly supports this goal.

The inter font which was sourced from Google Fonts was appropriate for the target audience due to it being modern and easily readable across all screen sizes. This is important due to the target audience being a wide range of people.

Bootstrap 5 was used as the CSS and the JavaScript framework. This was appropriate because it provided an adaptive layout which allowed the website to work on phones, tablets and desk computers. With that customers are able to browse on any device making this an important consideration.

How valid and reliable were the sources used?

The sources used in the project varied in validity and reliability. The most reliable sources were the Microsoft ASP.NET Core documentation at learn.microsoft.com and the Entity Framework Core documentation. These are first party sources that are maintained directly by Microsoft and is the definitive source for the technology used. These sources were crucial for the project because they provided accurate direction on controller patterns, migrations and identity authentication. Without these it would've been significantly more difficult to implement features such as role-based access control and the database structure.

The Google Fonts source from the inter font was also very reliable as its also a first party service which is operated and managed by Google with a well detailed and documented Open Font Licence that permits free commercial and non commercial use. Similarly,

Bootstrap 5 which is sourced from getbootstrap.com is a well trusted and well maintained open source framework used all over the industry.

Some of the images were less reliable due to its licensing clarity. For example, the images that were sourced from iStock and istockphoto.com require a paid license for commercial usage. The image that was used for the potatoes product was sourced from this platform and is used for educational purposes only. Similarly the image that was used for the chicken image that was sourced from boylesbutchers.ie was taken from a commercial retailer website and was not licensed for reuse. Had this not been a prototype but a properly deployed website these images would need to be replaced with properly licensed alternatives or an original image. However, the honey image sourced from Pixabay was a genuinely reliable source as Pixabay images are free for both commercial and non commercial use with no licensing required making this the most reliable image source.



Legal and ethical implication of using your assets

The legal implications for using the assets in the Greenfield Local Hub website were a very significant consideration. The project incorporates the use of several external sources, some of which carries licensing restrictions. Under the Copyright, Designs and Patents Act 1988, reproducing copyrighted images without permission or an appropriate license is illegal. For images that are sourced from iStock and commercial retailer websites such as boylesbutchers.ie, the use of these sources in this project was justified due to it being for only educational purposes. If the web application were to be deployed commercially then every image would need to be replaced with either original images of products, Pixabay licensed images or properly purchased stock images.

The inter font is licensed under the SIL Open Font License which allows free use in both commercial and on commercial projects. This means that there are no legal concerns with its usage. Bootstrap 5 is licensed under the MIT license also allows for unrestricted use. The Web Speech API used for the text to speech accessibility is browser native and carries no legal concerns.

From an ethical standpoint, following the law of GDPR in the handling of user data in the web application shows the ethical standards that were set out in the business proposal. The privacy policy page also included everything the user needs to know of how their data is handled which is ethically right.

The accessibility features that were implemented which includes dark mode, high contrast mode, text to speech and font size scaling all reflect an ethical commitment to inclusivity as required by the Equality Act 2010. By making the website usable for people with various kinds of accessibility needs, the web application avoids discriminating against users who would be excluded otherwise.

In the future I would handle the image sourcing differently by using only Pixabay or Unsplash images which are both free for commercial use or I would request for actual product images from the producers of Greenfield Local Hub. This would remove all copyright concerns and would also enable the product image to look more authentic and direct to the products being sold.

Project Outcomes

Overall, the Greenfield local Hub website prototype has successfully met the majority of its original requirements set out in the set task brief and business proposal. The platform provides customers with information about the producers, it allows them to view products with transparent pricing, place orders for delivery and collection, manage their accounts and view their order history. Producers are able to manage their stock levels, update product details and view incoming orders through a producer dashboard. The additional features that the client requested which includes a loyalty scheme, order management and scheduling as well as accessibility features were all implemented and are functional within the prototype. There are some areas where the prototype doesn't fully meet its potential. The Contact Us page does not currently send messages as it lacks a backend email handler. Order tracking feature for customers is limited to viewing a status label rather than receiving notifications in real time. Payment processing was not implemented as this would require a third party payment provider such as Stripe which was beyond the scope of the project. These limitations will be discussed in more detail in the improvement section.

Functional requirements

Users able to register and log in

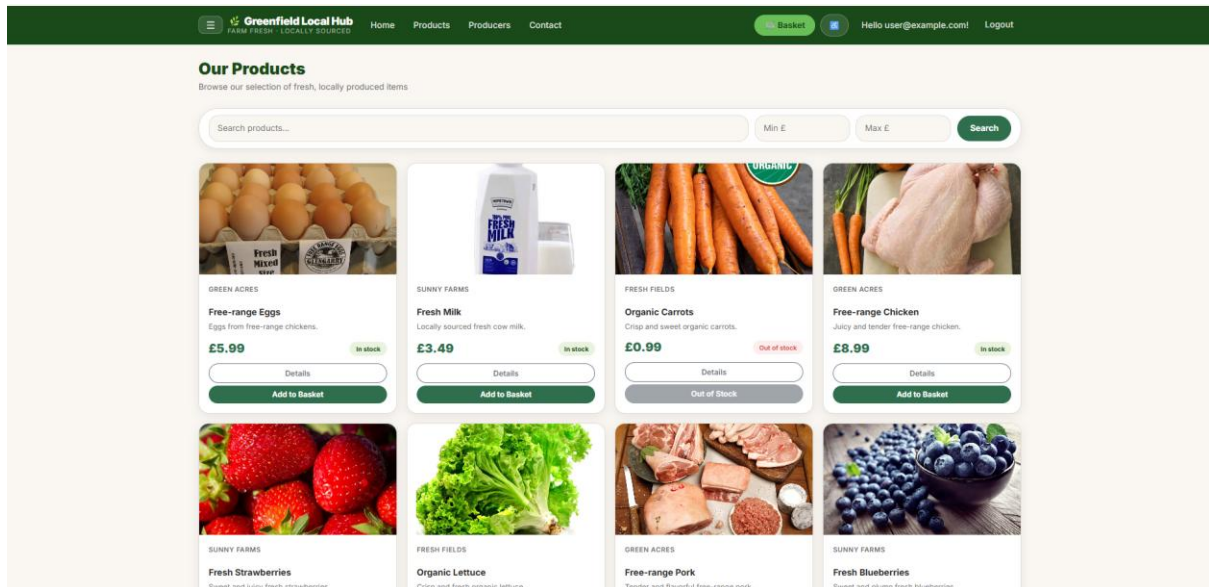
The prototype fully meets this requirement due to ASP.NET Identity being used to implement a complete authentication system which includes registration log in, log out and password management. The registration form validates email format and password and

displays meaningful error messages when there's an invalid input as confirmed by the test log. Feedback from the Observation feedback asking users to register an account showed that majority of the users were able to complete the task successfully with most finding the registration form easy to use.



Customers able to view products with transparent pricing and availability

This requirement was fully met. The products page displays every product available with its name, price, description farming method and a stock level badge indicating whether the product is in stock, low stock or Out of Stock. Feedback responses show that majority of users agreed that the product information was clear and detailed enough and that prices were displayed transparently. In the business proposal, it was stated that transparency about prices and availability from the start saves time of customers and the sales team and the products page successfully achieves this by making the information visible without requiring any customer service interaction



Customers able to place orders for collection and delivery

This requirement is fully met. The checkout process allows customers to choose between home delivery at a £3.99 or free with collection. A validation is enabled to ensure that delivery address is provided for home delivery and collection dates is also provided and must be 2 days in advance. Stock levels are checked before the order is confirmed to prevent overselling a product. Feedback from the scenario based questionnaire(Observation) showed that most users are able to place a delivery order successfully and felt confident they're order had gone through.

Checkout
Choose how you'd like to receive your order.

ORDER METHOD



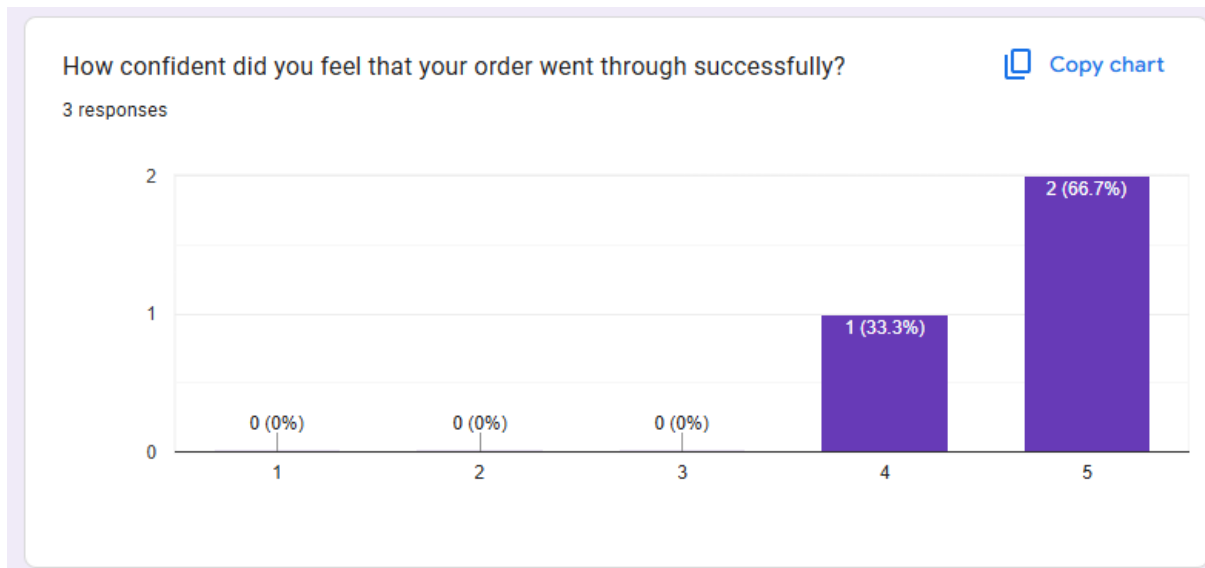
Home Delivery
£3.99 delivery fee



Click & Collect
Free — pick up from us

Place Order →

[← Back to Basket](#)



Customers able to manage their accounts and view order history

This requirement is partially met. Customers can view their full order history through the My Orders page which allows them to view all previous orders with their status, date, deliver method and total. They can also view the full details of each order which also includes all the products that were ordered. However, the ability to update account details such as email address and password through the website interface is limited to the default ASP.NET identity management pages rather than a custom built account management page. This could be improved in a future version by scaffolding the identity management pages and styling them to match the rest of the application.

Producer dashboard for stock, product and order management

This requirement is fully met. The producer dashboard displays stock levels for all of the producers products with colour coded badges indicating low stock, in stock, and out of stock. Producers can add new products, edit existing products, update stock levels and view incoming orders directly from the dashboard. The ability to update order status directly from the dashboard was also implemented following a review of the initial build. Feedback from the producer dashboard section of the questionnaire showed that user found the dashboard easy to navigate and the stock levels table clear and easy to update.

Order management, scheduling and tracking

This requirement is partially met. Users have the ability to choose a collection date or delivery date at the checkout . Producers and admins have the ability to update order status through the Edit Order page, which can be accessed through the Edit button visible to only Producers and admins. Customers can track the current status of their orders through the My Orders Page. However customers must check the website themselves rather than receiving automatic notification of an update in tracking status . In the future a feature for customers to receive email notifications on the change of their order status will be implemented to fully satisfy this requirement.

My Orders							
#17	24/04/2026	Shipped	Collection	26/04/2026	N/A	£1.88	Details
#18	24/04/2026	Ready for Collection	Collection	26/04/2026	N/A	£1.79	Details

Loyalty scheme

This requirement is fully met. The loyalty scheme automatically calculates a percentage discount based on the customers order history. Customers receive a 5% discount after one order, 10% after five orders and then 15% after ten orders. The discount is clearly shown in the basket along side the subtotal and final total. Feedback showed that majority of users were aware of the loyalty scheme after using the website and agreed that it encouraged them to return. The business proposal stated the scheme would entice new customers to become frequent customers. Due to the seamless nature of the discount method with no voucher codes, makes participation effortless.

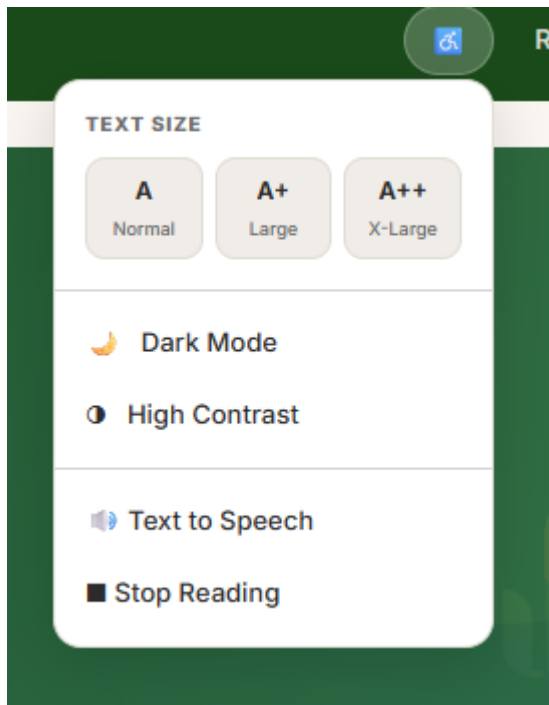
LOYALTY PROGRAMME

Earn points with every purchase
Get exclusive discounts — 5%, 10%, or 15% off based on your order history.

Sign Up Free →

Accessibility Features

This requirement is fully met and represents one of the best aspects of this prototype. The accessibility features implemented include dark mode, high contrast mode, three levels of font adjustment, a text to speech feature using Web Speech API, a skip link, ARIA labels on all interactive elements, visible keyboard focus indicators and a fully keyboard navigable interface. These features meet the WCAG 2.1 AA standard as required by the brief and Equality Act 2010. Feedback showed that users found the accessibility options easy to find and the features effectively.



Non-Functional requirements

Performance

In the business proposal, it was specified that webpages should load fully within 15 milliseconds and that 95% of pages should load within 3 seconds. During testing, the application ran locally using ASP.NET Core's development server and pages loaded quickly. Static assets such as CSS and images were served from wwwroot which is optimised for static file serving. Formal load testing against KPI of supporting 1000 users was not carried out as the prototype ran locally instead of being deployed to a production server. In the future when its being deployed this would need to be tested using a load testing tool such as Apache JMeter. Input delay was tested to be under 200 milliseconds across Chrome and Edge.

Security

In the business proposal, it was specified that all sensitive data should be encrypted using TLS 1.2 or higher and that user accounts should be protected using strong multi-factor authentication. In the system, passwords are hashed using ASP.NET Identity's built in PBKDF2 hashing algorithm and are never stored in plain text. Anti-forgery tokens are applied to all POST requests to protect against CSRF attacks. Role-based access control ensures that customers cannot access producer or admin functionality and that producers cannot edit another producer's products. Payment card data is not stored at all as no payment processing was implemented. A future version would integrate with a PCI-DSS compliant

payment provider such as Stripe to meet the payment security requirements specified in the proposal.

Usability and Accessibility

In the business proposal, it was specified that the system should maintain consistent design across all interfaces and that any page should be reachable within 3 clicks from the homepage. Both of these requirements are met in the system. The CSS design system using custom properties ensures a consistent visual identity across all pages. The navigation structure means that the products page, producers page, basket and orders are all accessible within 2 clicks from the homepage. The system meets the WCAG 2.1 AA compliance requirement through the diverse accessibility features that are described in the functional requirements section above. Feedback from the scenario based questionnaire form(Observation) showed that all 5 tasks could be completed in within a small number of clicks, with most users reaching their desired page within 1-2 clicks

Scalability

In the business proposal, it was highlighted that the system should support horizontal scaling without downtime. As the application was built using ASP.NET Core which is a stateless web framework, it is naturally scalable horizontally multiple instances could be run behind a load balancer. Entity Framework Core with SQL Server supports connection pooling which would help under high load. However the prototype was not deployed to a cloud platform so this scalability has not been formally tested. A future version deployed to Microsoft Azure using Azure SQL Database and Azure App Service would be able to demonstrate horizontal scaling with minimal configuration.

KPI- key Performance Indicators

The business proposal highlighted several KPI's across all the non functional requirements such as Performance, Security, Usability and capacity

Page Load Time: 95% of pages load within 3 seconds

This KPI was met during local testing. All pages of the web application loaded within 3 seconds during manual testing on a standard computer. A measurement for the 15 millisecond load target was not carried out due to it requiring production deployment and performance monitoring tools

Input Delay: Under 200 milliseconds

This KPI was met. The Ajax add to basket feature responds immediately with a toast notification all the form interactions were responsive throughout testing. There was no perceivable input delay observed when running on Chrome and Microsoft Edge

User Acceptance Criteria

Clementine and Ellie: Customer Personas

The user acceptance criteria for the customer personas Clementine and Ellie required that users could register, log in, view products with prices and availability, place orders for collection and delivery, and manage their accounts. All of these criteria were met by the prototype. The registration and log in pages are functional, validated and accessible. The products page displays all available products with transparent pricing, stock badges and filtering by search and price. The checkout flow supports both collection and delivery with appropriate validation on dates and addresses. Order history is also viewable on the My Orders Page.

The criteria also specified that all pages should meet WCAG 2.1AA standards. This was met through the diverse accessibility features that were implemented which includes dark mode, high contrast, text to speech, font size scaling and keyboard navigation. The feedback from the scenario based questionnaire(Observation) supports this as users with a range of technical abilities were able to complete all 5 tasks successfully.

An area where the customer acceptance criteria was only partially met was the account management. The criteria highlighted that users should be able to change their password and update their personal information. Though ASP.NET Identity pages meets the function, the pages were not fully customised to match the visual design of the overall web application.

Connor: Producer Persona

The user acceptance criteria for the producer persona Connor required that producers could log in, navigate to their dashboard, view and update stocks, edit product details and view incoming orders. All of these criteria were met. The producer dashboard displays all of the producer's products in a stock levels table with edit buttons, shows low stock counts and lists recent incoming orders with the ability to update their status. The criteria also specified that the Producer should be able to double check for accuracy before saving and the edit form fulfils this criteria by allowing producers to review all the fields before saving.

This acceptance criteria for Connor also required that producers could navigate to the order section and view full details of orders. This is met through the Orders link in the side bar which shows all orders containing the producer's products which is filtered by roles. Producers can then click details on any order to see the full breakdown of products ordered.

Future Improvements

Functional Contact Form

Currently the contact form on the Contact Us page doesn't send messages as there is no backend email handler connected to it. Implementation of services such as EmailJs or integrating an SMTP client in the home Controller would make this feature fully functional. This would benefit the customer by giving them a reliable way to get in touch with GLH directly from the website thus improving the level of customer service the business can offer. For the target audience which includes both young and older customers, having a clear and functional contact method essential for building trust.

Email Notifications For Order Status Changes

Currently customers must actively visit the web application to see if their order has been updated. So by adding automatic email notifications when an order status changes the user experience of the customer would be improved. This also aligns well with the order tracking KPI in the business proposal.

Payment Processing

Currently the prototype does not include payment processing. The integration of a PCI-DSS compliant payment provider such as Stripe would allow the platform to be deployed commercially. The business proposal specifically referenced the need for PCI-DSS compliant tokenisation to protect customer payment data. Implementing this would also require adding a payment confirmation step to the checkout flow and a receipt email to the customer thus further improving the overall user experience.

Product Reviews and Ratings

Adding the ability to leave star ratings and written reviews on products customers has purchased would be implemented in the future. This would increase the trust customers have for products and would provide valuable feedback for producers. This is very relevant to the target audience of local community members because they are more likely to trust recommendations from people than from the company themselves.

Customised Account Management Pages

Currently the identity management pages for changing passwords and email address use the default ASP.NET Identity UI which does not match the visual design of the rest of the application. In the future these pages would be scaffolded and the GLH CSS design system would be applied to create a fully consistent user experience throughout.

Allergen Filtering

Currently the products page allows users to filter by price and search term but not Allergen information . Customers may have serious allergies so an allergen filter would be ethically important and practical. The application already displays allergen information so implementing a filter wouldn't be difficult and it would be a significant improvement to the web application.